

Scheduling Algorithms OSSP

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FCFS Scheduling algorithm

```
#include <stdio.h>

int main() {
    int n, i;
    int bt[20], wt[20], tat[20];

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter burst time for each process:\n");

    for (i = 0; i < n; i++) {
        printf("P%d: ", i + 1);
        scanf("%d", &bt[i]);
    }

    // Waiting time
    wt[0] = 0;

    for (i = 1; i < n; i++) {
        wt[i] = wt[i - 1] + bt[i - 1];
    }

    // Turnaround time
    for (i = 0; i < n; i++) {
        tat[i] = wt[i] + bt[i];
    }

    printf("\nProcess\tBT\tWT\tTAT\n");

    for (i = 0; i < n; i++) {
        printf("P%d\t%d\t%d\t%d\n",
            i + 1, bt[i], wt[i], tat[i]);
    }

    return 0;
}
```

```
P4      5      10      21
monish_putti@monhh:~$ nano fcfs.c
monish_putti@monhh:~$ gcc fcfs.c -o fcfs
monish_putti@monhh:~$ ./fcfs
Enter number of processes: 4
Enter burst time:
P1: 8
P2: 4
P3: 9
P4: 5

Process BT      WT      TAT
P1      8        0        8
P2      4        8       12
P3      9       12       21
P4      5       21       26
monish_putti@monhh:~$
```

SJF_NONpreemptive

```

#include <stdio.h>

int main() {
    int n, i, j;
    int bt[20], wt[20], tat[20];
    int temp;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter burst time:\n");

    for (i = 0; i < n; i++) {
        printf("P%d: ", i + 1);
        scanf("%d", &bt[i]);
    }

    // Sort burst times
    for (i = 0; i < n - 1; i++) {
        for (j = i + 1; j < n; j++) {
            if (bt[i] > bt[j]) {
                temp = bt[i];
                bt[i] = bt[j];
                bt[j] = temp;
            }
        }
    }

    // Waiting time
    wt[0] = 0;

    for (i = 1; i < n; i++) {
        wt[i] = wt[i - 1] + bt[i - 1];
    }

    // Turnaround time
    for (i = 0; i < n; i++) {
        tat[i] = wt[i] + bt[i];
    }

    printf("\nProcess\tBT\tWT\tTAT\n");

    for (i = 0; i < n; i++) {
        printf("P%d\t%d\t%d\t%d\n",
            i + 1, bt[i], wt[i], tat[i]);
    }

    return 0;
}

```

```

monish_putti@monhh:~$ nano sjf.c
monish_putti@monhh:~$ gcc sjf.c -o sjf
monish_putti@monhh:~$ ./sjf
Enter number of processes: 4
Enter burst time:
P1: 8
P2: 4
P3: 9
P4: 5

Process BT      WT      TAT
P1      4        0        4
P2      5        4        9
P3      8        9       17
P4      9       17       26
monish_putti@monhh:~$

```

Priority

```
#include <stdio.h>
```

```
int main() {
```

```

int n, i, j;
int bt[20], priority[20], wt[20], tat[20];
int temp;

printf("Enter number of processes: ");
scanf("%d", &n);

printf("Enter burst time and priority:\n");

for (i = 0; i < n; i++) {
    printf("P%d Burst Time: ", i + 1);
    scanf("%d", &bt[i]);

    printf("P%d Priority: ", i + 1);
    scanf("%d", &priority[i]);
}

// Sort according to priority
for (i = 0; i < n - 1; i++) {
    for (j = i + 1; j < n; j++) {

        if (priority[i] > priority[j]) {

            temp = priority[i];
            priority[i] = priority[j];
            priority[j] = temp;

            temp = bt[i];
            bt[i] = bt[j];
            bt[j] = temp;
        }
    }
}

// Waiting time
wt[0] = 0;

for (i = 1; i < n; i++) {
    wt[i] = wt[i - 1] + bt[i - 1];
}

// Turnaround time
for (i = 0; i < n; i++) {
    tat[i] = wt[i] + bt[i];
}

printf("\nProcess\tBT\tPriority\tWT\tTAT\n");

for (i = 0; i < n; i++) {
    printf("P%d\t%d\t%d\t%d\t%d\n",
        i + 1, bt[i], priority[i], wt[i], tat[i]);
}

return 0;
}

```

```
monish_putti@monhh: ~  
monish_putti@monhh:~$ nano priority.c  
monish_putti@monhh:~$ gcc priority.c -o priority  
monish_putti@monhh:~$ ./priority  
Enter number of processes: 4  
Enter burst time and priority:  
P1 Burst Time: 8  
P1 Priority: 2  
P2 Burst Time: 4  
P2 Priority: 1  
P3 Burst Time: 9  
P3 Priority: 4  
P4 Burst Time: 5  
P4 Priority: 3  
  
Process BT      Priority      WT      TAT  
P1      4        1          0        4  
P2      8        2          4       12  
P3      5        3         12       17  
P4      9        4         17       26  
monish_putti@monhh:~$ _
```

ROUND ROBIN

```
#include <stdio.h>  
  
int main() {  
    int n, i;  
    int bt[20], rem_bt[20];  
    int wt[20], tat[20];  
    int quantum;  
    int time = 0;  
    int done;  
  
    printf("Enter number of processes: ");  
    scanf("%d", &n);  
  
    printf("Enter burst time:\n");  
  
    for (i = 0; i < n; i++) {  
        printf("P%d: ", i + 1);  
        scanf("%d", &bt[i]);  
  
        rem_bt[i] = bt[i];  
        wt[i] = 0;  
    }  
  
    printf("Enter time quantum: ");  
    scanf("%d", &quantum);  
  
    // Round Robin  
    while (1) {  
        done = 1;
```

```

    for (i = 0; i < n; i++) {
        if (rem_bt[i] > 0) {
            done = 0;

            if (rem_bt[i] > quantum) {
                time = time + quantum;
                rem_bt[i] = rem_bt[i] - quantum;
            }
            else {
                time = time + rem_bt[i];

                wt[i] = time - bt[i];

                rem_bt[i] = 0;
            }
        }
    }

    if (done == 1)
        break;
}

// Turnaround time
for (i = 0; i < n; i++) {
    tat[i] = wt[i] + bt[i];
}

printf("\nProcess\tBT\tWT\tTAT\n");

for (i = 0; i < n; i++) {
    printf("P%d\t%d\t%d\t%d\n",
        i + 1, bt[i], wt[i], tat[i]);
}

return 0;
}

```

```
monish_putti@monhh:~$ nano round_robin.c
monish_putti@monhh:~$ gcc round_robin.c -o round_robin
monish_putti@monhh:~$ ./round_robin
Enter number of processes: 4
Enter burst time:
P1: 8
P2: 4
P3: 9
P4: 5
Enter time quantum: 3

Process BT      WT      TAT
P1      8      15      23
P2      4      12      16
P3      9      17      26
P4      5      16      21
monish_putti@monhh:~$
```